

| Question number | Answer                                                                                           | Additional guidance                                                                                   | Mark       |
|-----------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------|
| 7(a)            | <ul style="list-style-type: none"> <li>change in the base {sequence / order} (of DNA)</li> </ul> | <b>ACCEPT</b> change in the number of chromosomes / damage to chromosome / loss of part of chromosome | <b>(1)</b> |

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|-----------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------|
| 7(b)(i)         | <ul style="list-style-type: none"> <li>a change in one variable is reflected in a change in another variable</li> </ul> | <b>ACCEPT</b> a description that only mentions 'change' once<br><b>DO NOT ACCEPT</b> causes / results in | <b>(1)</b> |

| Question number | Answer                                                                                                                                                                                                                                                                                                                         | Additional guidance                                                                                                                                                            | Mark       |
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| 7(b)(ii)        | <p>A description that includes the following points:</p> <ul style="list-style-type: none"> <li>incidence of skin cancer increases with (an increase in) age (1)</li> <li>incidence of skin cancer increases with an increase in the years (1)</li> <li>males have higher incidence of skin cancer than females (1)</li> </ul> | <p><b>ACCEPT</b> converse throughout</p> <p><b>ACCEPT</b> in either context of males or females or both</p> <p><b>ACCEPT</b> in either context of males or females or both</p> | <b>(3)</b> |

| Question number  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                    | Additional guidance                                                                                                                                                                                                                                                                                                | Mark       |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>7(b)(iii)</b> | <p>An answer that includes three of the following points:</p> <ul style="list-style-type: none"> <li>• as age increase there is more time for mutations to {build up / occur} (1)</li> <li>• people are spending more time in the sun (1)</li> <li>• more UV light reaching the earth's surface (1)</li> <li>• more men have {jobs / hobbies} outside (1)</li> <li>• men have a greater genetic disposition to skin cancer (1)</li> </ul> | <p><b>ACCEPT</b> the older you are the more exposure to risk factors / the more cell divisions the more chance of mutations</p> <p><b>ACCEPT</b> an example of why this may happen e.g. going on more holidays in hot countries</p> <p><b>ACCEPT</b> ozone layer depletion</p> <p><b>ACCEPT</b> men smoke more</p> | <b>(3)</b> |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| *7(c)           | <div> <p>Indicative content:</p> <p>Repeatability:</p> <ul style="list-style-type: none"> <li>sample size is small</li> <li>smokers are being studied and their sample size is very small*</li> <li>no statistical data available</li> <li>no indication of the location of the individuals (could be a validity comment as well)</li> </ul> <p>Validity:</p> <ul style="list-style-type: none"> <li>mean age at diagnosis is similar but not identical</li> <li>there is a greater range of ages for males than females</li> <li>no indication of the actual ages of the people</li> <li>so there could be more people at the extremes in one group</li> <li>no information about other lifestyle factors</li> <li>e.g. working in polluted environment, living in the city</li> <li>no information about non-lifestyle factors</li> <li>e.g. ethnicity</li> <li>no indication if individuals lived with other smokers (passive smokers)</li> <li>no indication of how long people been smoking for*</li> <li>no indication of how many cigarettes were smoked by the smokers each day*</li> <li>no indication if the non-smokers had ever smoked previously*</li> <li>no indication of severity of emphysema*</li> <li>no indication if emphysema is self-diagnosed or clinical diagnosis*</li> </ul> </div> <div> <p><b>Level 1 : comments on the design of the study</b></p> <p>1 mark = 1 comment<br/>2 marks = 3 comments</p> <p><b>Level 2 : comments on the design of the study</b></p> <p>3 marks = 4 comments, at least one of which is from each category<br/>4 marks = 5 comments, at least one of which is from each category</p> <p><b>Level 3 : comments on the design of the study and makes at least one link correctly with repeatability or validity</b></p> <p>5 marks = 5 comments at least one of which is linked to repeatability or validity<br/>6 marks = 6 comments at least one of which is linked to repeatability or validity <b>and</b> includes a specific comment about {smoking / emphysema}*<br/><br/><b>ACCEPT</b> reproducible / reliable<br/><b>IGNORE</b> accurate / precise</p> </div> |